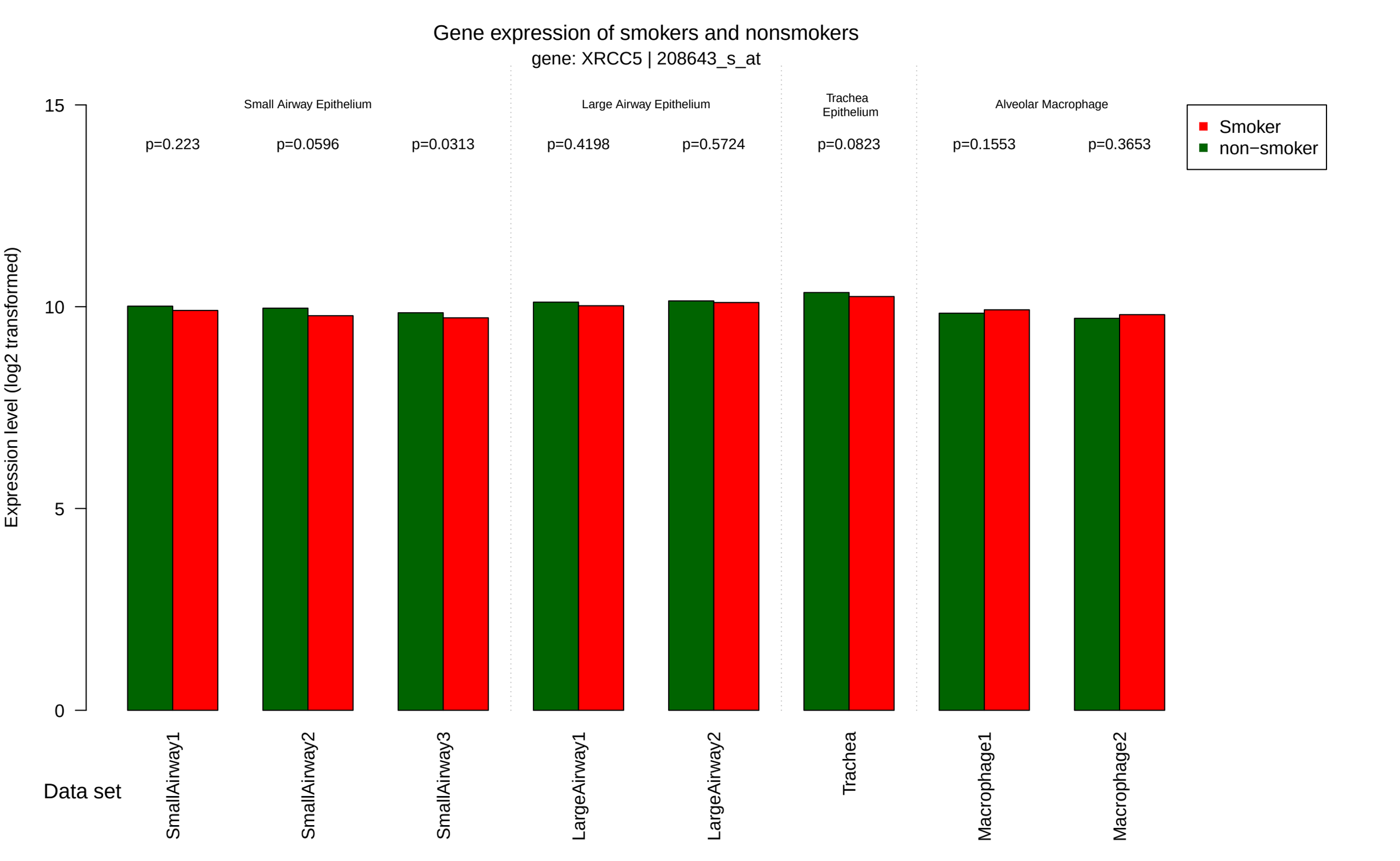
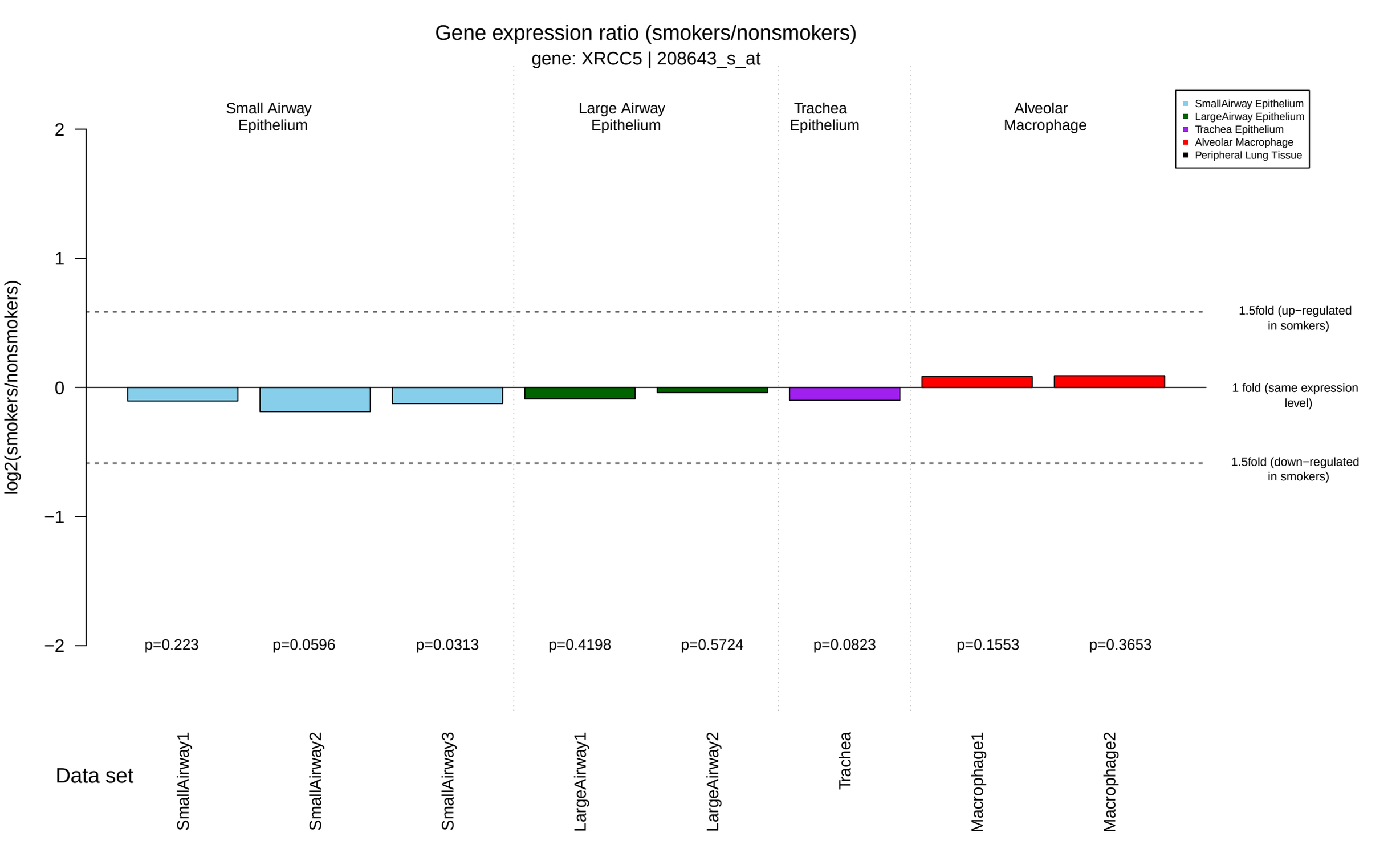
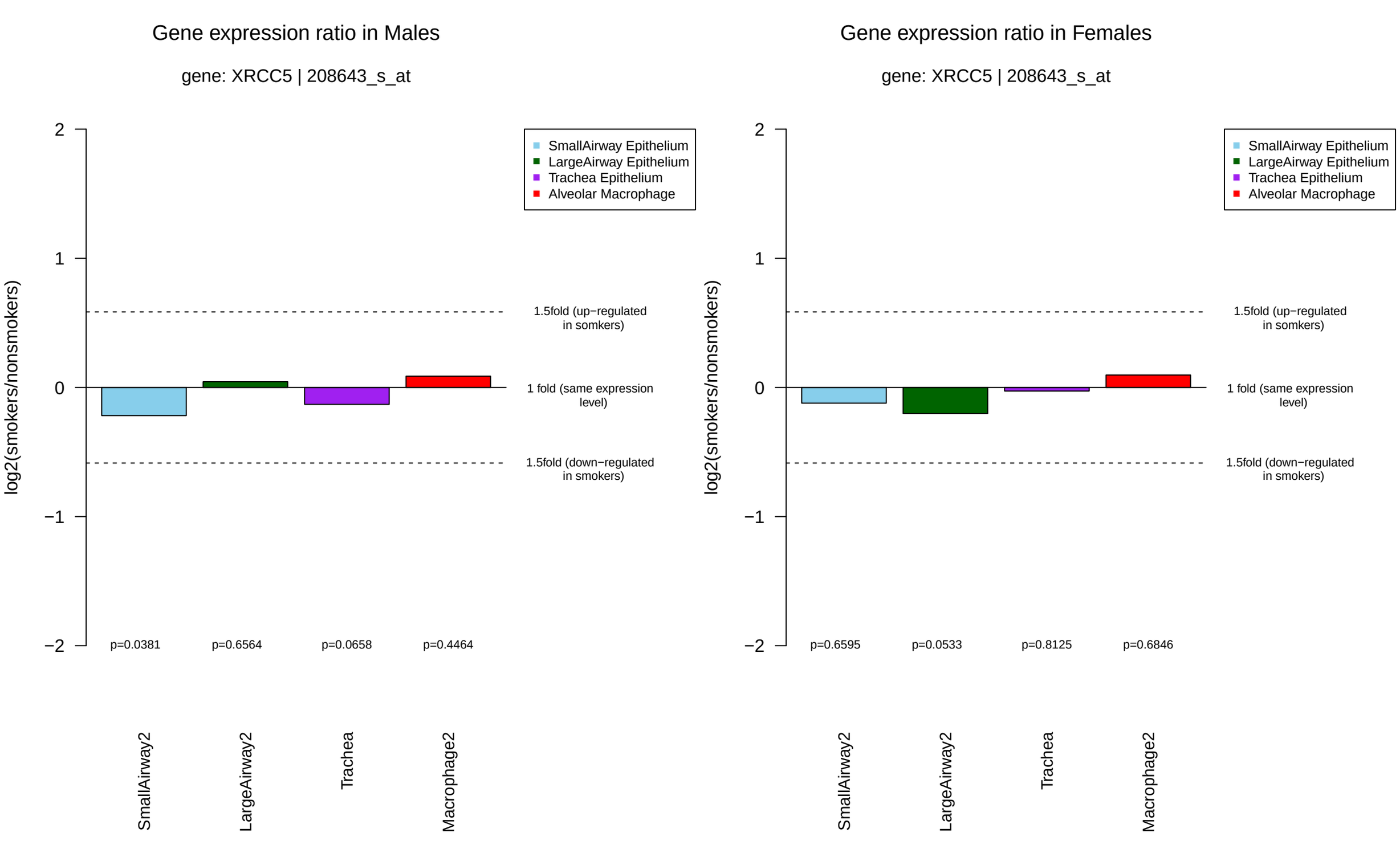
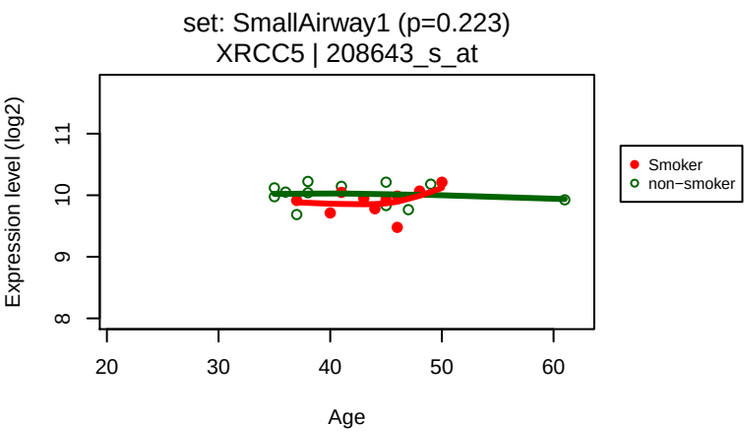
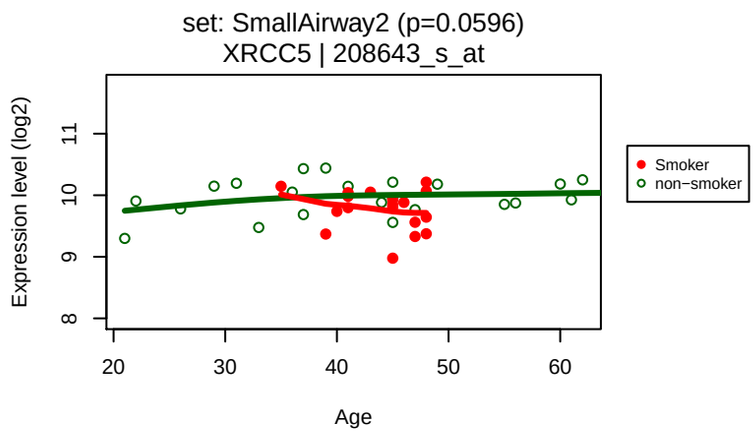
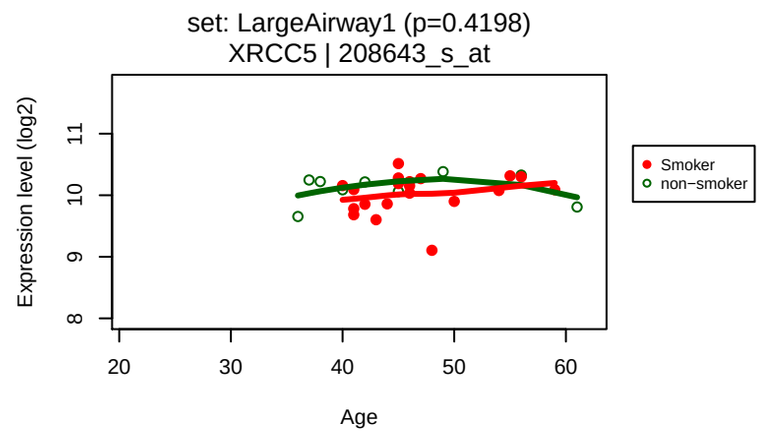

Gene expression of smokers and nonsmokers
gene: XRCC5 | 208643_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.223
p=0.0596
p=0.0313
p=0.4198
p=0.5724
p=0.0823
p=0.1553
p=0.3653
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

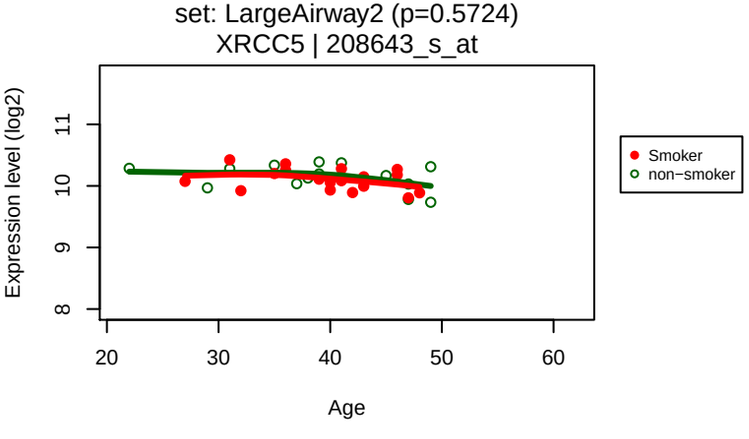

Gene expression ratio (smokers/nonsmokers)
gene: XRCC5 | 208643_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.223
p=0.0596
p=0.0313
p=0.4198
p=0.5724
p=0.0823
p=0.1553
p=0.3653
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

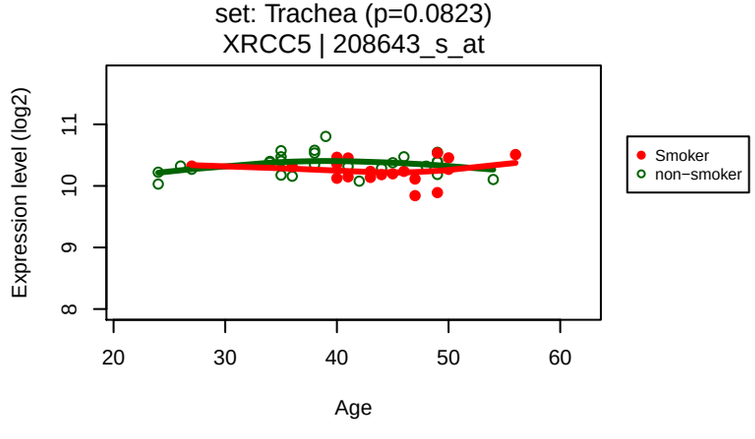

Gene expression ratio in Males
gene: XRCC5 | 208643_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0381
p=0.6564
p=0.0658
p=0.4464
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: XRCC5 | 208643_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6595
p=0.0533
p=0.8125
p=0.6846
SmallAirway2
LargeAirway2
Trachea
Macrophage2

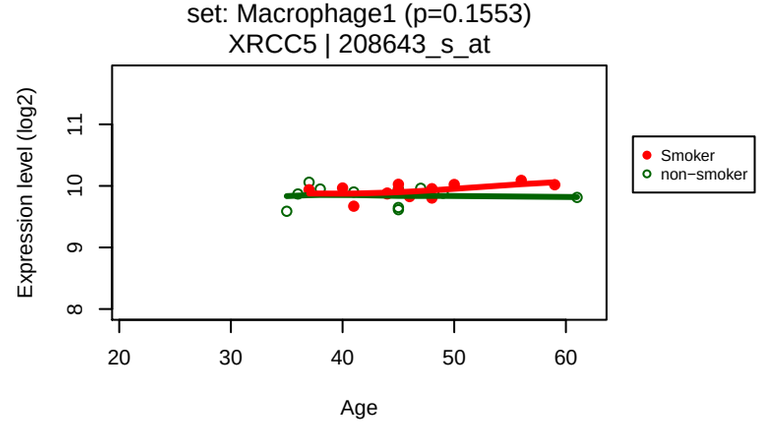

set: SmallAirway1 (p=0.223)
XRCC5 | 208643_s_at
Expression level (log2)
Age
Smoker
non−smoker

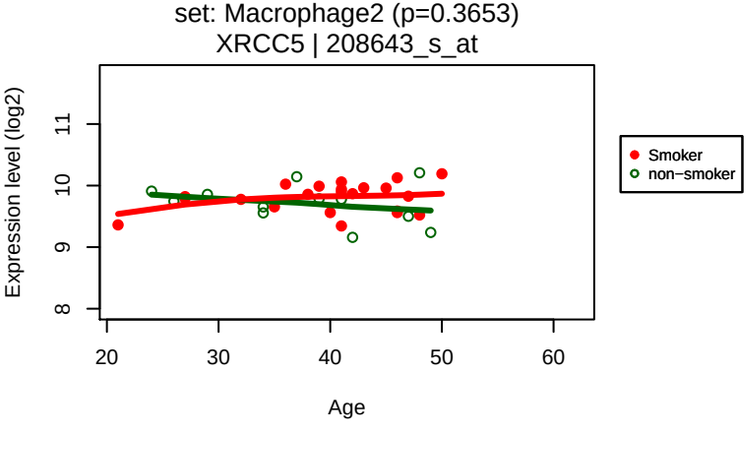

set: SmallAirway2 (p=0.0596)
XRCC5 | 208643_s_at
Expression level (log2)
Age
Smoker
non−smoker

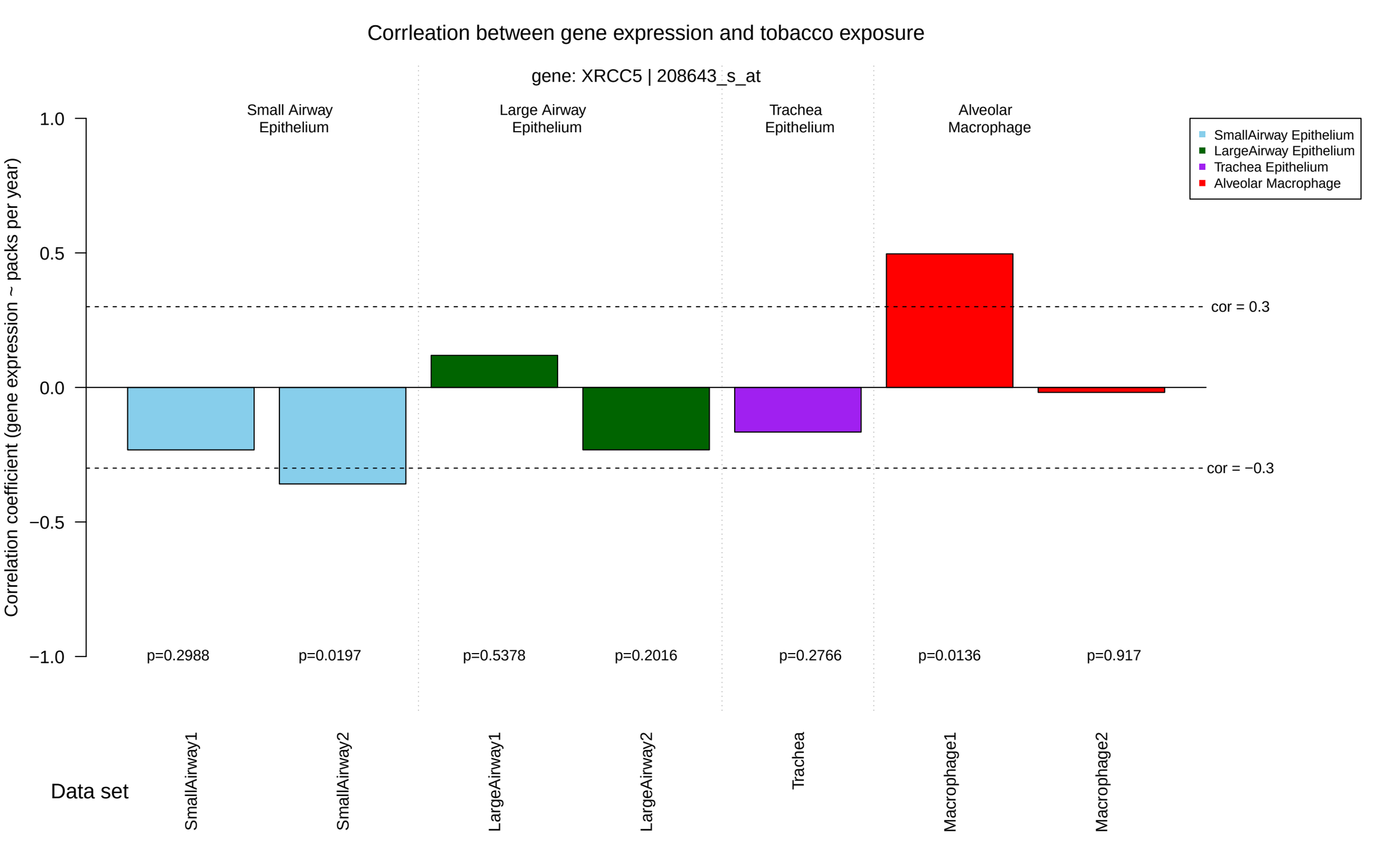

set: LargeAirway1 (p=0.4198)
XRCC5 | 208643_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5724)
XRCC5 | 208643_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0823)
XRCC5 | 208643_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1553)
XRCC5 | 208643_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3653)
XRCC5 | 208643_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: XRCC5 | 208643_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2988
p=0.0197
p=0.5378
p=0.2016
p=0.2766
p=0.0136
p=0.917
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2